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STRUCTURAL BIOLOGY

Exploring structural diversity across the protein universe with The Encyclopedia of Domains

Andy M. Lau†, Nicola Bordin†, Shaun M. Kandathil, Ian Sillitoe, Vaishali P. Waman, Jude Wells, Christine A. Orengo*, David T. Jones*

INTRODUCTION: The AlphaFold Protein Structure Database (AFDB) represents a major achievement in computational biology, containing high-quality predictions for more than 214 million protein structures. These structures are composed of domains, independently folding units that are critical to understanding protein function and evolution. The vast scale of the AFDB makes it difficult to accurately identify and classify these domains, thus limiting our ability to fully exploit its potential for biological discovery.

RATIONALE: To address this challenge, we developed The Encyclopedia of Domains (TED), a comprehensive resource for the systematic identification and classification of protein domains detectable within the AFDB. Traditional sequence-based methods, such as those used by Pfam and Gene3D, are unable to detect distant evolutionary relationships or entirely novel domains, but TED integrates multiple deep learning and advanced structural comparison techniques to uncover a much broader range of domain architectures, providing a more complete map of protein structure diversity.

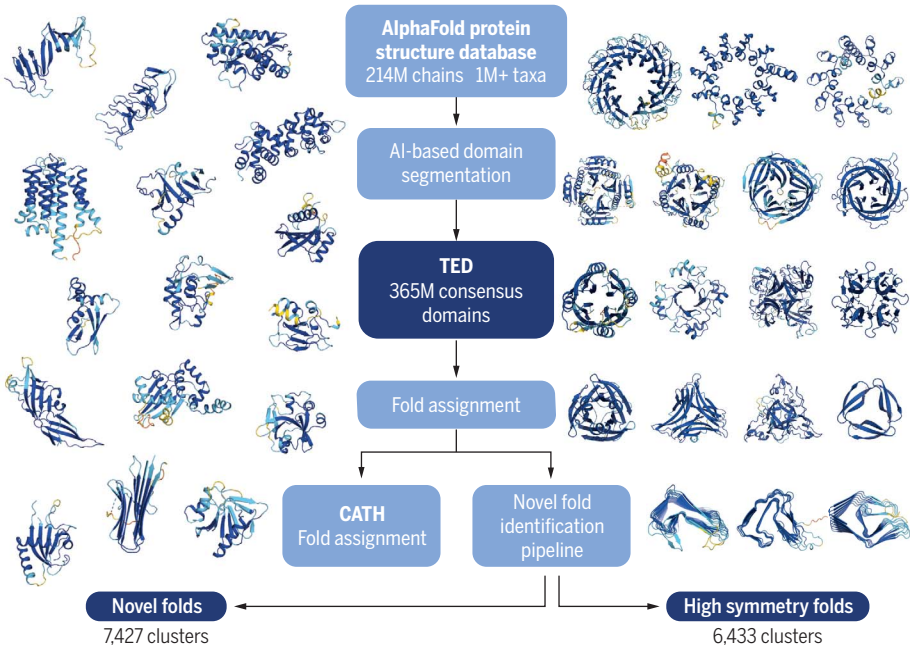
RESULTS: TED identifies nearly 365 million domains across more than more than 1 million taxa, substantially expanding the scope of domain detection in the AFDB. This includes more than 100 million domains that were previously undetected by sequence-based methods, showcasing TED's ability to leverage structure to increase domain coverage. We found that 77% of nonredundant domains were similar to existing CATH domains, validating the effectiveness of our approach.

Our study further highlights previously unobserved interactions between domains in the AFDB, uncovering more than 10,000 new structural interactions between superfamilies. We assessed the diversity of domain-packing geometries in interactions common to TED and CATH, and found that they were for the most part similar, with only around 5.4% of interactions showing a greater variety in TED. Overall, TED expands the set of known domain superfamily interactions to about three times the number seen in experimental structures, providing a means to greatly enrich our understanding of how multidomain proteins have evolved.

Moreover, TED has revealed thousands of putative new folds, substantially expanding the known fold space. This includes uncovering previously uncharacterized architectures, providing valuable new targets for structural and functional analysis. TED's analysis of domain distribution across the Tree of Life has uncovered patterns of fold exclusivity and shared architectures among different lineages, suggesting important evolutionary trends. For example, certain folds were found to be highly conserved across all superkingdoms, indicating their essential roles in cellular functions, whereas others were lineage specific, pointing to unique evolutionary adaptations.

TED facilitates a deeper understanding of the structural diversity within the AFDB by identifying highly symmetric and repetitive domain architectures, which are often associated with specialized biological functions. Previously unknown domains, including unique beta-propeller and alpha-helical structures, represent new opportunities for studying protein evolution and function. Additionally, TED has demonstrated the ability to detect new variations in domain packing, further enriching the structural data available in the AFDB.

CONCLUSION: TED is a resource that not only fills critical gaps left by previous sequence-based methods, but also opens up new avenues for research in structural and evolutionary biology. It will be continually updated to keep in step with future releases of the AFDB. The comprehensive domain annotations provided by TED will facilitate a wide range of functional analyses, from drug discovery to the study of protein evolution. By revealing previously hidden relationships within the vast protein structure space, TED sets the stage for future breakthroughs in understanding the complex interplay among protein structure, function, and evolution. ■



TED workflow. TED consists of 365 million domains identified from the AFDB by taking a consensus from three state-of-the-art domain segmentation methods. Nonredundant domains are assigned with CATH labels, and unassigned domains are subjected to a novel fold identification pipeline. Overall, we found several thousand novel folds, as well as novel high-symmetry fold clusters.

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Exploring structural diversity across the protein universe with The Encyclopedia of Domains

Andy M. Lau[1]†‡, Nicola Bordin[2]†, Shaun M. Kandathil[1], Ian Sillitoe[2], Vaishali P. Waman[2], Jude Wells[2,3], Christine A. Orengo[2]*, David T. Jones[1,2]*

The AlphaFold Protein Structure Database (AFDB) contains more than 214 million predicted protein structures composed of domains, which are independently folding units found in multiple structural and functional contexts. Identifying domains can enable many functional and evolutionary analyses but has remained challenging because of the sheer scale of the data. Using deep learning methods, we have detected and classified every domain in the AFDB, producing The Encyclopedia of Domains. We detected nearly 365 million domains, over 100 million more than can be found by sequence methods, covering more than 1 million taxa. Reassuringly, 77% of the nonredundant domains are similar to known superfamilies, greatly expanding representation of their domain space. We uncovered more than 10,000 new structural interactions between superfamilies and thousands of new folds across the fold space continuum.

The AlphaFold Protein Structure Database (AFDB) (*1*, *2*) is a groundbreaking initiative that has broadened the protein structure universe by expanding three-dimensional representation to >214 million UniProt sequences. The implications of the AFDB have been profound, not only in academic research in the life sciences, but also in the commercial sphere, where the integration of data and the advanced technologies used to generate accurate structures are being explored for next-generation structure-based drug discovery (*3*).

Notwithstanding its revolutionary impact, the AFDB is not without its limitations and presents a new set of challenges. The sheer scale of the data makes many traditional tools and pipelines that were originally designed for considerably smaller datasets inadequate for navigating the extensive number of structures and sequences within. This situation necessitates an evolved strategy to represent and traverse the data and complex relationships within such an expansive database, and requires synergy between algorithmic methods and computational hardware. Recent studies explored AFDB by partitioning full-length AFDB models into clusters of structurally similar proteins and characterizing their functions (*4*, *5*). At a more granular level, Bordin *et al.* (*6*) and Schaeffer *et al.* (*7*) interrogated the composition of specific proteomes (the initial AFDB 21 model organism dataset and 48 proteomes, respectively), cataloging domains under the CATH (*8*, *9*) and ECOD (*10*) frameworks.

Domain discovery is possible through sequence- and structure-based approaches, with Pfam (*11*, *12*) and Gene3D (*13*) as prime examples of the former. The Pfam database describes collections of protein families, each represented by a multiple sequence alignment (MSA) and a hidden Markov model (HMM) profile (*11*, *12*). In Gene3D, sequences of existing CATH superfamilies, assigned through structure, are leveraged to discover new domains in sequence space through representative profile HMMs (*13*). Sequence-based discovery allows for greater coverage but is limited by the detection capabilities of HMMs, which often fail on remote relatives. Structure-based assignments allow for higher-quality domain boundaries and can reveal very remote relatives, but have been limited by the low numbers of experimental structures. The task is further complicated by conceptual difficulties when performing structural comparisons, such as the lack of a statistical null model (compared with the use of random sequences for sequence comparisons).

An essential perspective yet to be explored in the context of the AFDB is the comprehensive mapping and analysis of protein domains across various branches of the Tree of Life using structural data. The relationships between protein folds and domains have been extensively highlighted by structural databases such as CATH (*8*, *14*, *15*), ECOD (*10*), SCOP (*16*), and SCOPe (*17*), with differences in criteria for which regions are assigned to domains in proteins. Although early comparisons across domain databases showed agreement for fold-level assignments (*10*, *18*, *19*), differences exist for the definition of a protein domain. CATH recognizes that some structures can be decomposed into further structural units, each with their own repertoire of observed variations, whereas SCOP takes into account the idea of fold recurrence, i.e., whether a subunit has been observed as reoccurring in another family or only as a single domain (*18*).

Examination of the AFDB through the lens of the CATH framework could potentially illuminate subtle yet important connections between structure and function across large swathes of organisms. As such, a structure-based mapping of domains within the AFDB promises to facilitate the exploration of these relationships and provide a foundation for further annotation.

In this study, we present a comprehensive analysis of domain composition for the entirety of the AFDB (version 4). This description encompasses 365 million putative domains derived from >214 million protein sequences across >1 million taxa. The identification of these domain structures is made through the consensus of three automated parsing methodologies: Merizo (*20*), Chainsaw (*21*), and UniDoc (*22*). Further, by using structural comparison methods such as Foldseek (*23*) and an in-house deep learning method called Merizo-search, >251 million domains can be placed on the CATH hierarchy.

A high-throughput strategy for identifying structural domains in the AFDB

Our workflow combines three state-of-the-art domain parsing methods together with structure classification algorithms to identify known domain folds (Fig. 1, A and B, and see the materials and methods) within the AFDB. Using this workflow, we identified a total of 365 million "TED" domains across the AFDB (fig. S2 and table S1), 100 million more domains than found through sequence-based methods (Fig. 1C). TED-100 describes an ~42:55 division between single and multidomain proteins (Fig. 1, Di), with the latter up to 20 domains in composition (fig. S3). Only 2.8% of targets (5.3 million) in TED-100 lack identifiable domains, compared with 33.9% (64.1 million targets) in Gene3D and 26.2% in Pfam (49.4 million targets) (fig. S4). In TED, these targets either consist entirely of nondomain residues (NDRs) or lack any consensus among the three domain segmentation methods used. Across superkingdoms, the fraction of NDRs varies, with ~10% identified across archaeal and bacterial lineages and up to 30% in eukaryotes (fig. S5).

Analysis of the average predicted local distance difference test (plDDT) scores for TED-100 reveals that the majority of domains fall within the "very high" to "high" bins, with only ~2% within the lowest bin (Fig. 1Dii). Because our domain segmentation methods do not consider residue plDDT when determining domains,

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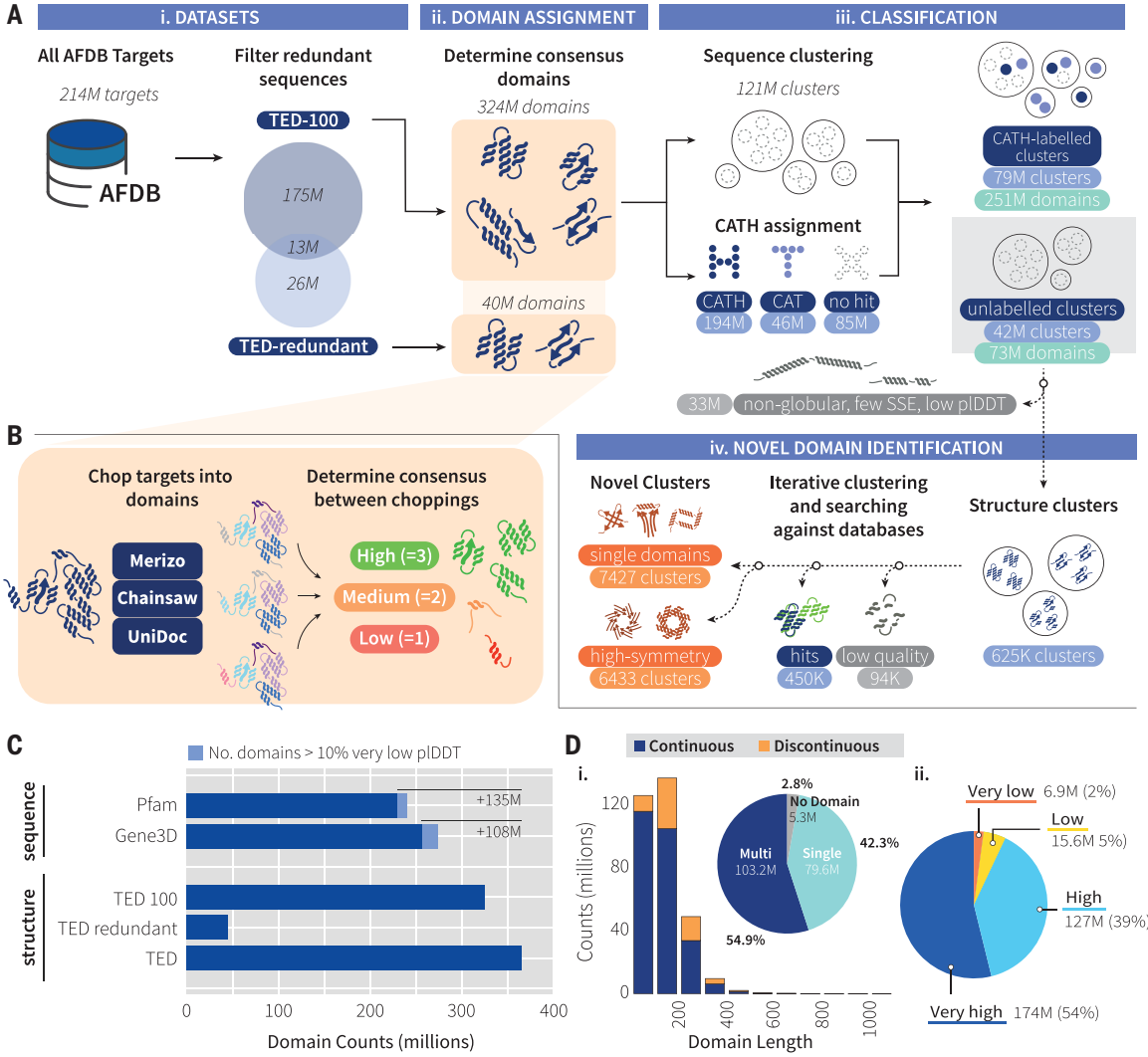

Fig. 1. Overall workflow. (**A**) (i) A total of 214 million AFDB target sequences are filtered by 100% sequence identity to avoid bias. This identifies 188 million nonredundant targets (TED-100) and a set of sequence-redundant targets (TED-redundant). (ii) Both TED-100 and TED-redundant undergo automated domain parsing, with assignments derived from consensus among the three methods. (iii) TED-100 domains are processed by MMseqs2, creating >121 million clusters at 50% identity. Concurrently, domains are matched to CATH domains through Foldseek and Merizo-search, categorized into superfamily (C.A.T.H.), topology (C.A.T.), or no matches. Domains found by Merizo-search nearest-neighbor matches are considered as topology-level matches. Clusters are annotated with CATH labels, creating partially labeled and unlabeled clusters. Low-quality domains in unlabeled clusters are filtered out. (iv) Resultant domains undergo a new workflow for identification involving clustering and database searches for matches to known structures. Poor-quality domains (non–protein-like) are identified using an in-house deep learning method (see the materials and methods). Novel domains are additionally scored on internal symmetry using the SymD program (see the materials and methods). (**B**) Full-length targets are subjected to automated domain parsing by Merizo, Chainsaw, and UniDoc. A consensus is taken by identifying assignments where three methods agree (high), two methods agree (medium), or no consensus is found (low). Only high- and medium-consensus domains are analyzed further. (**C**) Comparison of domains identified by sequence (Pfam and Gene3D) versus structure-based methods (TED). The "TED" count combines TED-100 and TED-redundant. (**D**) (i) Domain length distribution and proportion of identified continuous (blue) and discontinuous (orange) domains. Inset shows proportion of single domains, multidomains, and number of targets with no identified domains (n = 188,914,411). (ii) Average plDDT distribution for TED-100 domains (n = 324,389,697) across confidence bins: dark blue indicates very high confidence (plDDT ≥ 90); blue indicates high confidence (90 > plDDT ≥ 70); yellow indicates low confidence (70 > plDDT ≥ 50); and orange indicates very low confidence (plDDT < 50).

this suggests that our domain-finding pipeline effectively identifies plausible domains within the well-folded areas of AF2 models.

Classification of TED domains into the CATH hierarchy

The 324 million domains in TED-100 were clustered by sequence using MMseqs2 (*24*) and compared against CATH representative domains using fast structure searching methods (Fig. 1Aiii and see the materials and methods). This produced ~121 million clusters at 50% sequence identity and a minimum coverage of 90% (fig. S6 and see the materials and methods). The majority of these clusters comprise singleton sequences (~81 million), with the largest nonsingleton cluster comprising 12,847 domains.

Parallel to sequence clustering, we used a combination of Foldseek (*23*) and Merizo-search (*25*) (an in-house structure search method using domain coordinate embedding) to search all TED-100 domains against CATH SSG5 domains (*9*), allowing 194 million domains to be assigned with CATH superfamily (H) labels and 46 million at the topology (T) level (see the materials and methods). These labels were further validated by scanning domain sequences against an updated library of HMMs for CATH Protein Data Bank (PDB) domains. Approximately 171 million superfamily predictions by Foldseek could be confirmed with HMM superfamily matches (88.54%), with an additional 1.8 million domains (0.95%) confirmable at the fold level. Only 4.1 million of the 16 million Foldseek predictions for fold matches on CATH can be validated by HMM scans (25.8%), with 11.8 million fold predictions and 20.3 million superfamily predictions by Foldseek not confirmed by an HMM match, suggesting an expansion by 15.4% in CATH-labeled domain coverage using AFDB structures over HMM-based sequence assignments.

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By identifying sequence clusters with any CATH-labeled domains, the clusters were partitioned into two categories: 78 million clusters (>251 million domains, 77% of all TED-100 domains) containing at least one CATH-labeled member, spanning 148 million proteins, and 26 million proteins having no domain annotations in Pfam and 30 million having none in Gene3D (fig. S7). The remaining 41 million clusters have no members with CATH labels (~73 million domains; fig. S6). The absence of similarity to CATH domains in the latter clusters could be attributed to them being novel folds, extremely divergent relatives of existing CATH domains, or simply incorrect models, making them unmatchable to known folds.

Enrichment of fold representation by TED

To develop an understanding of how folds are distributed across the AFDB, we assessed the composition of TED using the CATH hierarchy. Figure 2A shows the top 100 CATH superfamilies of each class (alpha, beta, and alpha/beta), which are greatly enriched in TED-100 compared with baseline sequence hits in Gene3D.

These folds and architectures are unevenly distributed across the Tree of Life; however, the majority of folds (61%) are shared and reused across all superkingdoms, suggesting essential roles. Some folds were found across two superkingdoms (18.5%), whereas others are more exclusive, with 0.5, 9, and 11% of CATH folds found only in archaea, eukarya, and bacteria, respectively.

The most abundant superfamilies in TED-100 and CATH are shown in figs. S8 and S9. Directly comparing the top superfamilies assigned in TED compared with CATH in terms of raw domain counts results in the promotion of several superfamilies into the top five of each class, including the MFS general substrate transporter–like domain, translation factors, and the FAD/NAD(P)–binding domain (fig. S8). The set of superfamilies highly enriched in TED include those associated with the archetypal multidrug efflux pump AcrB. AcrB is a resistance-nodulation-division (RND) transporter and forms part of the AcrAB-TolC efflux pump in bacteria, where it is responsible for the export of harmful substances such as antibiotics, thus contributing to antibiotic resistance (*26*). The constituent domains of AcrB, including the pore domain, transmembrane domain, and TolC-docking domain, are found greatly enriched in TED compared with the PDB, providing up to an ~1000-fold increase in representation for these biologically important superfamilies.

The pore domain, which forms part of the pore selectivity filter in RND transporters, is principally found in bacterial species and in a small minority of archaeal and eukaryotic lineages based on Gene3D hits. However, structure-based searching through TED expands the coverage of the pore domain superfamily into an additional 18 archaeal, 1315 bacterial, and 284 eukaryotic lineages unique to TED, which evaded even HMM searches. This broader coverage of organisms, revealed by structural comparisons, may reveal potential evolutionary events, such as lateral gene transfer between bacteria and eukaryotes. This expansion is exemplified in Fig. 2C, where we show that >1000 CATH superfamilies, described by Gene3D sequence matches as occurring solely in a single superkingdom, could actually be mapped to other branches when structure was taken into account in TED.

TED also enables us to better study the distribution of folds across different branches of the Tree of Life. Among the 193 million TED domains with superfamily labels, we observed folds that were exclusively localized to specific superkingdoms. These findings are shown in Fig. 2D, where folds are summarized at the CATH topology level as visualized by principal component analysis (PCA), and where points at the vertices represent exclusive occurrences of a topology and all of its superfamilies in a particular superkingdom.

High-symmetry architectures

From our TED workflow (Fig. 1), we identified 41 million sequence clusters that could not be linked to CATH superfamilies. Representatives of these clusters were subjected to a workflow aimed at identifying previously unknown domain folds (Fig. 1Aiv and see the materials and methods).

While reviewing these clusters, it was apparent that we would need to treat repeat architectures with high internal symmetry separately. A good example of domains in this class are the various WD40 beta propellers, which are considered distinct domain architectures in their own right, but clearly comprise repeats of domain-like units. To identify similar domains in our workflow, we calculated Z scores using the SymD program (*27*), sequestering any cluster representatives with a symmetry Z score of >9 into a category of 6433 highly symmetric novel fold clusters (fig. S10).

Within these clusters, we found architectures that have not been seen before, such as an 11-bladed beta-propeller, a closed alpha-ring-like 11-helix propeller, and a 6-helix propeller (Fig. 3). Various other propeller arrangements, including the "beta-flower" domain shown by Durairaj *et al.* (*4*), were also identified, with some other highly symmetric examples shown in Fig. 3.

We also found a broad category of architectures composed of cyclic repeats, extruded along an axis to form highly repetitive and symmetric structures which we call "extruded repeats." Several studies (*28*, *29*) and databases such as the Database of Structural Repeats in Proteins (DbStRiPs) (*30*) have curated repeat architectures from the PDB, including alpha- and beta-solenoids, and horseshoes formed of alpha-repeats such as HEAT, ankyrin, and armadillo. Figure 3 showcases some complex examples of these structures, with some featuring highly varied and unstructured loops between repeating units (fig. S11). Many of these domain architectures resemble other solenoid folds that were recently reported in a systematic study of beta-solenoid fold space (*29*).

Novel domains and their distribution across the Tree of Life

The remaining low-symmetry clusters were assessed on domain quality using a variant of the Foldclass network (*25*) trained to identify poor-quality domain choppings (see the materials and methods) and novelty by further matching against known structure libraries. The final output of our workflow produced 7427 clusters of putative domains, which appear to be well folded but dissimilar to any known domain fold. Although there is no exact boundary between novel and just highly divergent examples of known folds, by applying a density-based anomaly detection algorithm (see the materials and methods), we could at least rank these domains by novelty relative to known domains.

Figure 4 showcases examples of novel domains identified by our workflow. More than a quarter of clusters corresponded to singletons at the sequence-cluster level (1930 domains), which are uniformly distributed in terms of the Foldclass novelty score. Most novel domains identified are from bacterial proteins, which is unsurprising given that the latter comprise most proteins in the AFDB and in our TED-100 dataset. Overall, these domains are distributed evenly across different phyla, but several bacterial phyla were found overrepresented when compared with baseline counts across all domains in TED-100, specifically in the PVC group, including the Myxococcota, Spirochaetota, Bdellovibrionota, Nitrospinae/Tectomicrobia, and Calditrichota phyla (fig. S12), suggesting that species within these phyla may be underrepresented in terms of domain coverage in the current PDB.

Ranked first by novelty is a curious archaeal domain found as a sequence singleton from *Candiatus Poseidoniales archaeon* (TED: A0A7M3WA57_TED05; novelty score: 77.2; Fig. 4Ci). This protein is not documented in InterPro and no Gene Ontology (GO) terms are available. The structure is composed of paired beta-strands in a closed, twisted hairpin with both termini adjacent to one another. Reviewing the context of the full chain from the AFDB shows that the domain forms part of an extended loop protruding from the middle of a immunoglobulin-like domain (TED: A0A7M3WA57_TED04). The hairpin topology of A0A7M3WA57_TED05 is mirrored by another identified novel domain represented by E1Z635_TED02 shown in Fig. 4Dii, but is alphahelical in nature. This domain is found only in eukaryotes, primarily in species belonging

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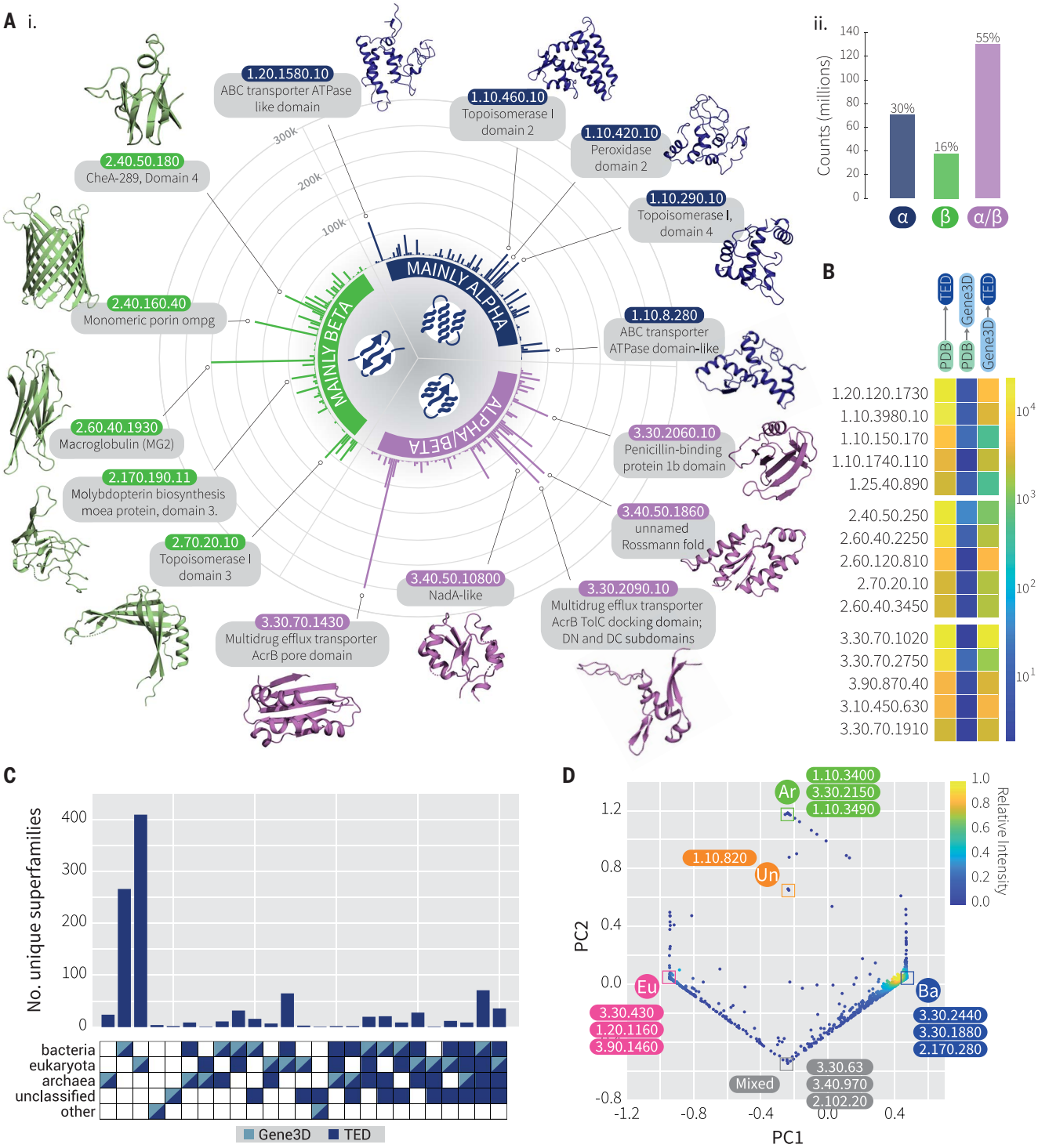

Fig. 2. Classification of TED domains using the CATH hierarchy. (**A**) (i) The top 100 superfamilies in TED-100 for each CATH class where more matches to CATH superfamilies have been identified through structural hits in TED compared with sequence hits in Gene3D. (ii) Proportion of domains matched to CATH classes (n = 238,569,631). (**B**) Enrichment of superfamily representation in TED-100 compared with PDB and Gene3D. The top five superfamilies of each CATH class are shown, where enrichment in TED-100 compared with PDB is the greatest. Color scale represents fold change in superfamily representation in PDB and Gene3D compared with Gene3D and TED. A full list of fold names corresponding to the CATH superfamily codes can be found in table S2. (**C**) Expansion of CATH superfamilies to new superkingdoms in TED. Plot shows the number of unique superfamilies found in each superkingdom (across the 653,460 taxa of TED-100) according to Gene3D and TED assignments. Each column along the horizontal axis depicts the number of superfamilies that are exclusive to a single superkingdom when only considering Gene3D assignments but are expanded into one, two or three additional superkingdoms in TED. Only superfamilies where Gene3D domains are exclusive to a single superkingdom are shown (n = 1061). (**D**) Exclusivity of CATH topologies across superkingdoms. PCA of normalized CATH topology counts across five superkingdoms: eukarya (Eu), bacteria (Ba), archaea (Ar), and unclassified and other sequences (Un). The "mixed" category comprises topologies found in approximately equal proportions in the Eu and Ba domains. Examples of superkingdom-exclusive topologies are shown for each category.

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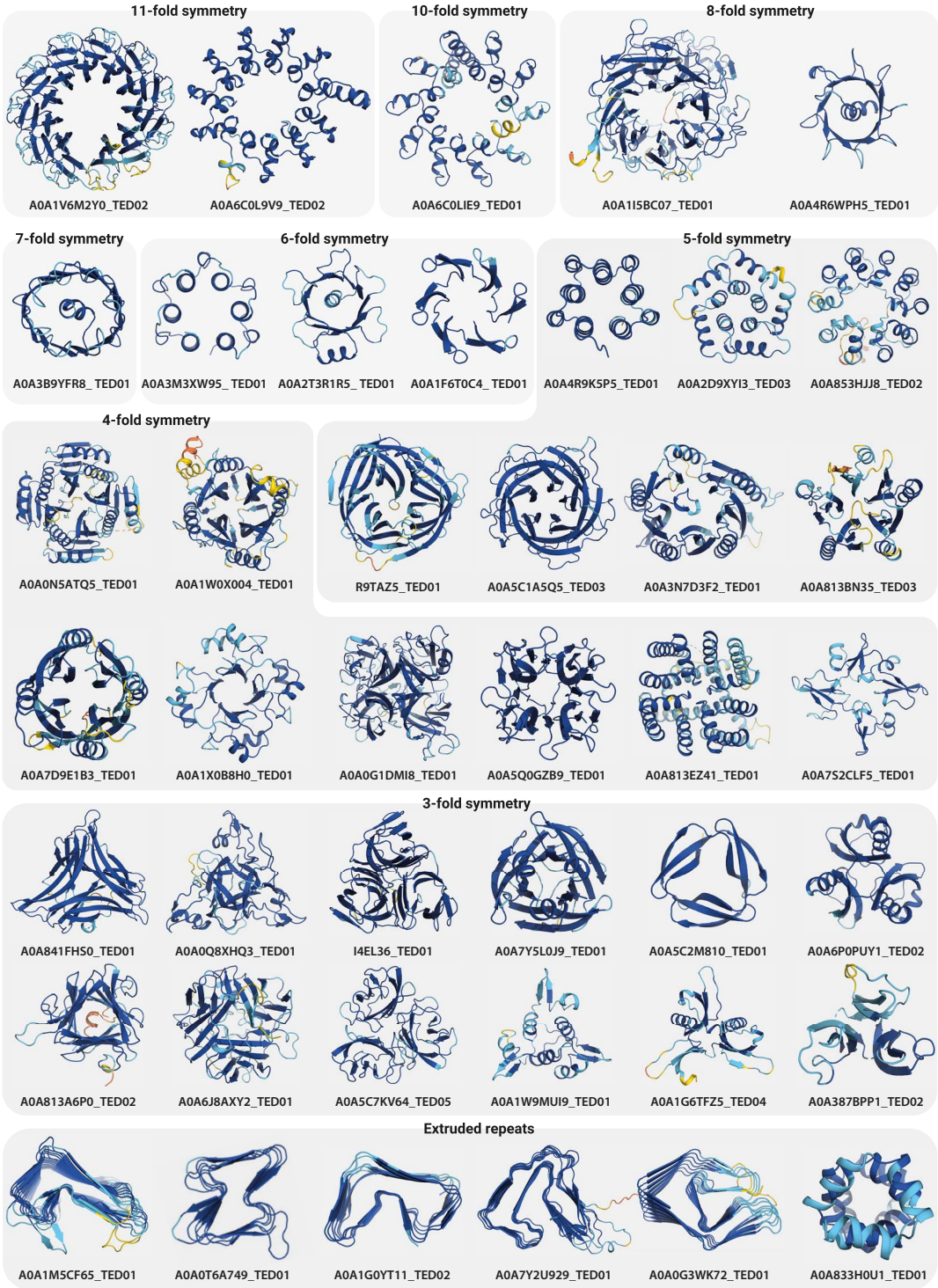

Fig. 3. Examples of high-symmetry domains and extruded repeats. Domains are identified as part of the novel domain identification pipeline and are identified as domains with high internal symmetry through scoring with the SymD program (see the materials and methods). Extruded repeats are domains with a high number of ordered cyclical repeats projecting along one axis. Coloration follows pIDDT confidence bins as per the AFDB. Dark blue indicates very high confidence: pIDDT ≥ 90; blue indicates high confidence (90 > pIDDT ≥ 70); yellow indicates low confidence (70 > pIDDT ≥ 50); and orange indicates very low confidence (pIDDT < 50).

to the Viridiplantae phyla (97 species) but also in a minority of Opisthokonta (14 species) and Amoebozoa (1 species), suggesting an evolutionary link between these lineages.

Sequence-based function prediction for novel fold and repeat domains

To determine whether any functions could be assigned to the domains with novel folds and repeats, we use a sequence-based deep learning model (see the materials and methods) to predict GO terms. This analysis shows that 1321/7427 (18%) of the domains in the putative novel fold set, and 1419/6433 (22%) of the repeat set can be assigned high confidence ($p < 10^{-4}$) molecular function GO term labels. The top 20 GO terms predicted for the two sets of domains are shown in table S4 and table S5. Manual inspection of the domains predicted to have zinc-binding and nucleic acid–binding functions reveals that many of the domains contain plausible zinc-binding sites (*31*), with most containing two Cys and two His residues arranged in a tetrahedral fashion, including as part of zinc finger–like supersecondary structure motifs. Figure 4Ei shows one example containing two zinc-binding sites and a possible

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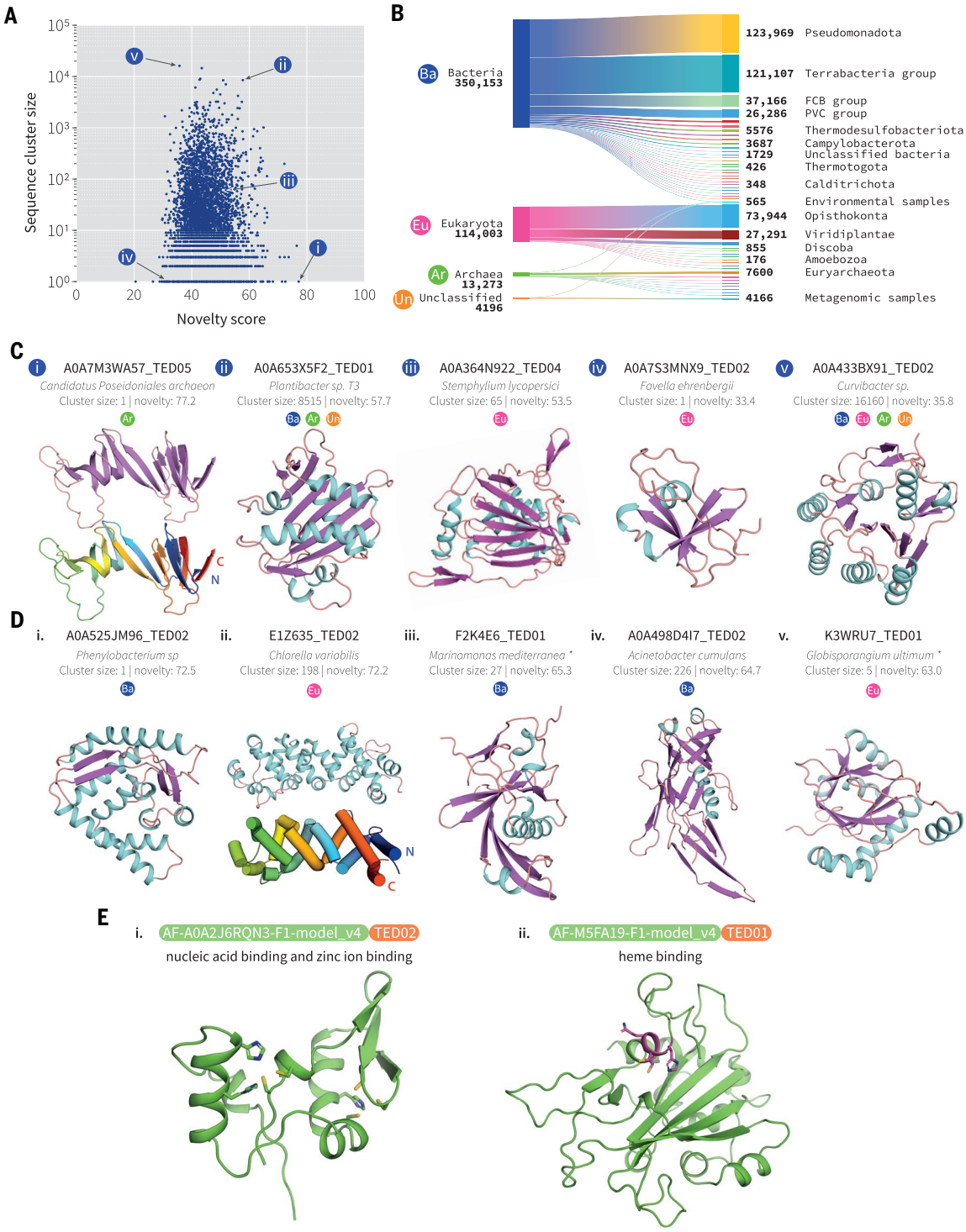

Fig. 4. Examples of novel domain clusters identified in TED. (**A**) Comparison of domain novelty score versus sequence cluster size (n = 7427). Novelty scores are predicted by the Foldclass algorithm where novel domains are ranked with a score close to 100. (**B**) Taxonomic distribution of novel domain clusters (for all sequence cluster members; n = 483,732). Largest common phyla are shown across superkingdoms, along with the number of domains in sequence clusters assigned to each level of the hierarchy. (**C**) (i) to (v) correspond to labels shown in (A). In (i), the bottom subpanel shows the arrangement of strands that form the coiled hairpin loop from the N terminus (blue) to the C terminus (red). The quoted cluster size represents the number of identified homologs at the sequence cluster level. Labels denoting superkingdoms correspond to (B) and represent the superkingdom to which all cluster members belong. The cluster is distributed across multiple superkingdoms when multiple labels are shown. (**D**) Examples of high-novelty structures. In (ii), the bottom subpanel shows the arrangement of helices that form the coiled hairpin loop from the N terminus (blue) to the C terminus (red). Asterisks denote where organism names have been shortened: (iii) *Marinomonas mediterranea* (strain ATCC 700492/JCM 21426/NBRC 103028/MMB-1), (v) *Globisporangium ultimum* (strain ATCC 200006/CBS 805.95/DAOM BR144) (*Pythium ultimum*). (**E**) Novel folds with predicted functions. (i) Example of a domain predicted to have nucleic acid and zinc binding properties. Potential zinc-binding site residues are highlighted as sticks. The left site is composed of two Cys and two His residues, whereas the right site has three Cys residues and one His residue in a tetrahedral arrangement. (ii) Example of a heme-binding domain. The residues of the heme c–binding motif are highlighted.

nucleic acid–binding alpha-helix, but which lacks the canonical zinc finger supersecondary motif. We also consider the set of domains predicted to have heme-binding properties and find that most of these contain the canonical heme c–binding motif (CXXCH) (*32*). Inspection of the three-dimensional structures reveals that each of these domains has one or more heme-binding sites in a plausible conformation; one example is shown in Fig. 4Eii, with the residues of the heme c–binding motif highlighted. The His residue that binds the heme iron is in a conformation compatible with placement of the heme group in the pocket, which is primarily hydrophobic. The presence

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of clear sequence motifs and structural features consistent with the assigned functions suggests that these novel domains may indeed have the predicted functions, and further work is needed to validate the remainder of the predicted functions for these novel domains.

Interactions between domain pairs

Unlike sequence-based domain annotations such as Gene3D, the availability of full-chain AF2 models for multidomain proteins allows us the unique opportunity to interrogate and compare domain pair–packing interactions in TED and CATH. TED contains a total of 27,280,057 instances of interacting domains categorized into 13,771 interacting superfamily pairs (ISPs). By contrast, the set of interacting domains in CATH consists of just 196,234 instances across 5111 ISPs. The relative enrichment in the number of instances of ISPs common to CATH and TED is shown in Fig. 5A, which shows that most of these ISPs have many more members in TED. Assessing the diversity of interaction geometry for ISPs common to TED and CATH (Fig. 5B, i and ii, and see the materials and methods), we find that for most ISPs, the diversity of interaction geometries in TED is consistent with that seen in CATH, indicating that on average, AF2 tends to recapitulate interdomain geometries already seen in the PDB.

A small proportion (5.4%) of these ISPs show enhanced diversity in interaction geometries in TED, as measured by an increase in the conservation of interaction orientation (CIO) score (see the materials and methods) of 0.3 or more. A smaller proportion of ISPs (2.4%) are more diverse in CATH. We find no strong dependence of difference in CIO value on whether the ISP in question is homotypic (two copies of domains from the same superfamily) or heterotypic (Welch two-sample t test, $P = 0.141$). Most of the ISPs in the TED set (10,701 out of 13,771) are unique to TED, i.e., they are not observed in CATH, and CATH contains 2041 ISPs not seen in TED (fig. S13).

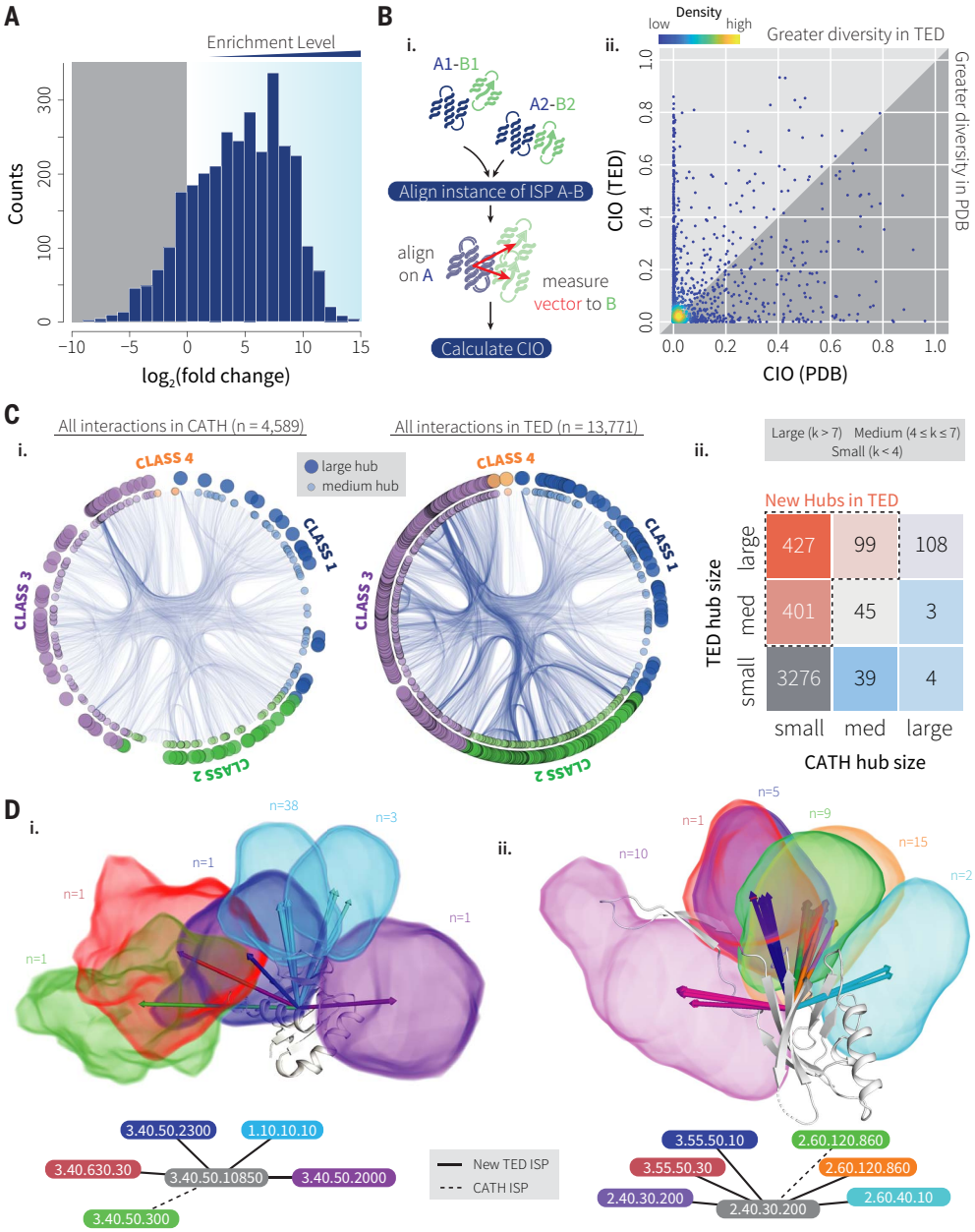

Fig. 5. ISPs. (**A**) Enrichment of the number of instances of ISPs common to the CATH and TED datasets expressed as $\log_2$(fold change) (n = 3070). (**B**) (i) Alignment procedure used to compute CIO values for an ISP. One domain in each instance of each ISP is used as a reference and aligned to a designated "master" reference domain structure. The rotation and translation from each alignment is applied to the second, "tag-along" domain to bring all domain pairs into a common frame. Vectors are then computed between the centers of mass of each pair of domains and used to compute the CIO measure (see the materials and methods). (ii) Comparison of CIO values for ISPs common to CATH and TED. Most ISPs show a high degree of conservation in interaction patterns. (**C**) (i) Hierarchical edge-bundling plots illustrating differences in domain superfamily interaction patterns between CATH (left) and TED (right). Curves in the plots connect interacting superfamilies. Hubs are marked by medium (four to seven connections) and large circles (more than seven connections) on the outer rim. (ii) Comparison of hub domains in CATH and TED. The heatmap compares CATH superfamilies in CATH and TED as hubs, categorized as small (fewer than four connections), medium (four to seven connections), and large (eight or more connections). Hub thresholds used are from Ekman *et al.* (*48*). (**D**) Two examples of new hub superfamilies in TED, with groups of domains for interacting superfamilies placed in a common frame and represented as volumes alongside chains involved in each group and a graphical representation of each hub. The sets of interactions for superfamily 3.40.50.10850 (from NtrC-like two-domain protein) are nonoverlapping between CATH and TED, whereas the set of TED interactions for superfamily 2.40.30.200 (distal tail protein domain) include that seen in CATH (orange and green). A decomposed view of the interactions for superfamily 3.40.50.10850 is shown in fig. S15.

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That some interactions in CATH are not captured in TED is not entirely surprising, because of the possibility of domain parsing and classification errors. Also, TED excludes any sequence not modeled in the AFDB, and these include sequences from viruses (~5,304,757 sequences) and longer sequences, which we estimate contain ~50 million domains (see the materials and methods). The AFDB will also not contain experimental constructs comprising domain combinations not observed in natural sequences. Finally, instances of ISPs in TED are filtered based on whether the constituent domains are in contact in the full-length AF2 structure and if they have a favorable interdomain predicted aligned error (PAE) (see the materials and methods), both of which depend on the availability of homologous sequences and template structures that ideally span both domains. For these reasons, we do not expect that TED ISPs will be a strict superset of CATH ISPs. Nevertheless, we find that TED essentially triples the set of known domain interactions at the superfamily level (10,701 novel TED interactions versus 5111 known interactions in CATH).

A visual illustration of ISP sets can be seen in Fig. 5Ci, in which a path is drawn between two superfamilies if at least one interaction is observed between domains assigned to those superfamilies. This representation uses the CATH hierarchy as a guide to "bundle" paths drawn between superfamilies in related parts of the CATH hierarchy (fig. S13 and see the materials and methods), and shows visually that a very large number of new interactions are seen, especially between superfamilies in CATH classes 2 and 3 (all-beta and alpha-beta classes, respectively). As mentioned above, the majority of the interactions seen in the TED set are unique to TED, and a visual comparison of the set of TED-unique interactions to all known interactions in CATH (fig. S13) reveals a huge expansion in the set of known interactions. The network of superfamily interactions also allows us to identify superfamilies that can be considered hubs because of their ability to interact with many other superfamilies. As shown in Fig. 5C, i and ii, a large number of superfamilies in CATH have their hub status "promoted" due to new interactions seen in TED. In Fig. 5D, i and ii, we show two examples of superfamilies that have been promoted in this way. The set of interactions observed in TED for these superfamilies show that they can contain only novel (in the case of 3.40.50.10850) or, in some cases, a mixture of novel and already-observed interaction modes (for 2.40.30.200).

The interaction data shows that TED greatly expands the set of known interactions and interaction modes between domains. Future work will aim to delve deeper into the interaction data and determine the functional roles that these interactions (particularly those not seen before) might play in cellular processes, in particular taking into account the multidomain architecture of individual proteins and their evolutionary relationships.

Structures of redundant sequences in the AFDB

Of the 214 million structures in the AFDB, nearly 39 million are exact sequence duplicates of other proteins in the database (13 million unique sequences within this set). The 26 million redundant proteins comprise our TED-redundant set (see the materials and methods).

The AFDB models for these sequence-redundant proteins often diverge from one another, with ~42% of clusters (5.6 million) showing a maximum cluster RMSD of >1 Å (fig. S16). The largest RMSDs in the distribution tend to relate to changes in domain packing, but even at the domain-fold level, changes can be observed. Figure S16 shows the distribution of maximum pairwise RMSD for each cluster of identical sequences. We found structural variation at the chain level (fig. S16A), as well as in the PAE maps generated by AF2 (fig. S16Aii).

One explanation for our observations here is that we are looking at alternative conformers of the protein chains relating to different MSAs. However, in these cases, the input MSAs should be identical given the described modeling protocol. The larger differences are also far too large to attribute to the short relaxation step that follows the model prediction step. This is most evident in the example shown in fig. S16Aii, where two AFDB models for identical sequences deviate by nearly 65 Å and show clear differences in the PAE map.

To further investigate structural diversity within these sequence-redundant clusters, we subjected the TED-redundant targets to our domain parsing workflow, identifying just over 40 million domains across the set. In addition, we derived a consensus across all structures in each identical sequence cluster (see the materials and methods). This allowed us to investigate domain-level changes in conformation between identical sequences and to identify many cases where the consensus domains were very different (fig. S16B).

Discussion and conclusions

What we have shown so far in developing TED is a way in which structural data in the AFDB can be augmented by carefully breaking down structures into their component domains, allowing them to be classified through the CATH framework. This initiative not only drives forward the associations that we can make between structure and function, but, as shown in our study, can be used to discover and reclaim the dark areas of fold space that are not accessible to sequence-based discovery.

Comparing TED with a recent study on the 21 model organisms dataset (*6*) (released before the 214 million release), already shows that the TED workflow not only identifies a greater number of domains, but also that the domains are of higher quality and capture many more remote homologies (table S3). TED currently annotates domains for >1 million taxa, of which 600,000 are currently mapped to CATH domains in TED-100, and such an extended mapping of domains to superfamilies will enable many more evolutionary discoveries. A good example of how such expansion of CATH superfamilies enhances understanding of evolutionary processes and aids inheritance of functional properties, even toward drug repurposing, is shown in fig. S17.

The coverage of TED dwarfs sequence-based assignment methods (namely Gene3D and Pfam), identifying >100 million more domains in the AFDB compared with the latter. The proportion of proteins that each method can find domains within is shown in fig. S4, which illustrates the advantage of considering structural domains (TED finds domains in 40 to 50 million proteins that Gene3D and Pfam cannot). Prior studies based purely on sequence comparisons have suggested that between 40 and 65% of prokaryotic proteins are composed of multiple domains, with a higher proportion proposed in eukaryotes (*33*–*35*). These values are comparable to those seen in TED, where we find an ~42:55% split between single and multidomain proteins in the AFDB (fig. S4). Compared with TED, the proportion of multidomain proteins is much lower in Gene3D (29%) and Pfam (24%) assignments (fig. S4).

Although most structures in the AFDB are undeniably of high quality, the sheer scale of the data means that errors and anomalies are inevitable, and these should be discussed to assess the overall robustness of the data. Some limitations have already been pointed out by other studies (*36*, *37*). One such idiosyncrasy that appeared during our development of TED was the observation that models of 100% identical sequences were sometimes substantially different. To reframe the issue, the implication of this is that for a given protein, a user may be able to find a better model or domain within the AFDB (and one that AF2 clearly places higher confidence in) if sequence-redundant copies are considered. This may mean that some alternative structures could only be detected when redundant copies are available to compare against. Given that the most AFDB entries (175 of 214 million) do not have duplicates, the implication here is that an unknown (but probably large) number of low-quality structures or regions in AFDB might be improvable by resampling the input MSAs and reassessing the quality of the models. Even though this might be too heavy a task to do across the whole of AFDB, users of the database should certainly bear this in mind when looking at specific domains of interest.

The obvious explanation for the existence of divergent models for a given sequence must be

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that the models were generated using different MSA information, and we note that explicit presampling of the MSAs given to AF2 has been explored recently as a way to persuade the network to generate alternate conformations (38, 39). Because the structure generation pipeline in AF2 includes possible random MSA resampling during the forward pass, it could be that a borderline MSA could result in very divergent predictions on different runs due to this resampling (fixed random seeds were not used in making the AFDB). Another possibility is that the sequence data banks used for different instances of the same target sequence were inadvertently updated to newer versions if they were predicted at later times. We were not able to investigate these possibilities further ourselves because there is no information on the MSAs provided in the AFDB data, but we do suggest that it is a topic worthy of further investigation by the AFDB developers.

Fortunately, we find many examples that reiterate the notion that plDDT is an invaluable discriminator of the confidence that a user should place on a model. Figure S18 shows an example whereby an AFDB entry has been generated within the very-low-confidence bin. This model features structural defects not typical of AF2 models. "Refolding" the sequence in ColabFold (40) produces a number of very different structures across the five AF2 models, the variation of which, along with their unusually low plDDT scores, strongly indicates that AF2 has hallucinated these folds and that they should likely be disregarded.

Overall, however, the large proportion of domains mapping to CATH evolutionary families gives confidence in the quality of the models, capturing well the structural features of folds and preserving their distinctive structural characteristics. In this context, the performance of the domain segmentation algorithms deployed here has also been key, because early pilot work on the 21 model organisms suggested a much higher proportion of problematic AF2 models largely caused by the poor segmentation of full-length proteins into domains by the sequence-based methods. In addition to confirming earlier hypotheses that the majority of domain structural families had already been characterized experimentally (41, 42), our study has also revealed some intriguing and beautiful new domain architectures and folds, especially some highly symmetric repetitive structures. Future development of TED will aim to comprehensively analyze these new folds and incorporate them into the CATH hierarchy, which will require extensive manual curation. Additionally, we recognize the need to optimize the current TED workflow for better detection of repeat structures, such as the different extruded repeats highlighted in our work. This optimization will necessitate the development and integration of new tools and analysis methods specifically designed to identify these types of structures.

Throughout our study, we had to make a number of algorithmic decisions that were primarily motivated by the monumental amount of data we had to process in the AFDB. Therefore, we intend for TED to be an ongoing development that will evolve as the data and the needs of its users do. Our aim is to provide the community with the most comprehensive summary and breakdown of the structures within the AFDB. We expect TED to be used as a starting point for a whole host of analyses, including providing a comprehensive dataset that can be used to train and test a new generation of deep learning–based applications in structural biology.

Materials and methods

Datasets

Our analysis was performed on version 4 of the AFDB, containing models for 214,683,829 UniProt sequences. To avoid overrepresentation bias, most of our initial analysis was performed on a nonredundant subset of 188,914,411 sequence unique AFDB models (TED-100). There are 38,944,835 targets that are sequence redundant, and 13,175,417 are in TED-100 (sequence representatives were selected as the first target in a sorted list). The remaining 25,769,418 are unique to the TED-redundant dataset.

Deriving a unified assignment of domains in the AFDB

Both the TED-100 and TED-redundant datasets were subjected to a consensus domain parsing workflow that made use of three segmentation methods, Merizo, Chainsaw, and UniDoc, before deriving a consensus between their outputs. Each method is provided with the structure of the AFDB model and returns the predicted domain ranges. In the case of UniDoc, which does not classify NDRs, models were first parsed using Merizo to remove NDRs before subsequent segmentation into domains with UniDoc. This two-step procedure for generating UniDoc domains, in which NDRs are first removed, is necessary because otherwise the presence of NDRs can mislead UniDoc into classifying full chains as single domains. Furthermore, removal of NDRs through using only plDDT scores is insufficient, as previously shown (20). Overall, 400,444,974, 328,956,414, and 366,117,430 putative domain regions were identified by Merizo, Chainsaw, and UniDoc, respectively (table S1 and fig. S2), requiring several months of computation using the CS Cluster at UCL. Computer hardware consisted of a mixture of both central processing units (CPUs; ~400 cores at peak) and graphics processing units (GPUs; up to ~100 NVIDIA GTX 1080 Ti or RTX 2080 Ti). GPUs were used for deep-learning based methods Merizo and Chainsaw, whereas UniDoc was run using only CPUs. Further details on how consensus domains were derived are described below.

Domain classification using Foldseek

Foldseek (23) (version a435618b95ba95cbfe74dcbc8b4bd2720547d285) easy-search was benchmarked to assess tentative homology and fold assignments thresholds on a curated set of CATH domains. We created a dataset of 3186 domains representative of CATH classes 1 to 4 clustered at 30% sequence identity with equivalent superfamily and boundary assignments in agreement with SCOP. Using a genetic algorithm, we identified thresholds for superfamily (H-level) and fold (T-level) level matches in CATH at a 98% precision level with the following parameters: E-value cutoff of 0.108662, minimum coverage of 0.366757, coverage mode 5 (shortest sequence is at least 36.6757% of the longest sequence), and sensitivity 10. Further postprocessing on the raw results from Foldseek was applied by using custom thresholds for H-level and T-level hits. At this precision level, the recall at the H and T levels is ~0.59 and 0.71.

We selected as valid H-level hits any results that passed the thresholds set to Foldseek, with an additional cut-off based on a template modeling (TM) score >0.56, a coverage threshold of 0.367, and an E-value <0.019. Hits at the fold level (T) required different cutoffs, with results considered as valid hits if passing the Foldseek E-value cutoff, a coverage cutoff of 0.786, and a minimum TM score of 0.42. Using these thresholds, we scanned all 324,389,697 domains with a medium to high confidence against a library of 31,574 CATH SSG5 (structural similarity groups 5 Å, where all CATH superfamily members are within 5 Å when superposed) (9), resulting in 193,939,494 hits at the H level and 16,026,530 hits at the fold level, leaving 114,423,673 domains with no labels assigned at this stage.

Domain classification using embedding similarity

To further reduce the number of unmatched domains, we subjected the remaining domains not matched by Foldseek to an in-house structure-embedding search method (Merizo-search) to identify further matches to our CATH SSG5 representative dataset. Merizo-search uses a deep learning method called Foldclass that makes use of an equivariant graph neural network to encode a domain structure into a fixed-size embedding (25). A query domain is embedded using Foldclass and then compared against a database of CATH domain embeddings, with similarity determined through cosine distance between the target and CATH domain embeddings. To further verify the top match, TM-align (43) is used to align the query coordinates against the CATH domain corresponding to the closest domain embedding. The nearest-neighbor CATH domain is considered a hit if the TED domain

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achieves a TM-align score >0.5 (normalized by the length of the TED domain).

Sequence clustering of TED-100 domains

The sequences of TED-100 domains were clustered using MMseqs2 (version 22a77e22a77eeb1b6c64f1c5f04e1480b4705bb3fbc897), enforcing a minimum sequence identity of 50% and at least 90% coverage of the shorter sequence against the longer sequence (coverage mode 5). Sequence clusters were then populated using CATH labels identified using Foldseek and Foldclass (25) methods (see sections "Domain classification using Foldseek" and "Domain classification using embedding similarity" above). A full breakdown of the number of labeled and unlabeled sequence clusters and the number of domains comprising these clusters is available as table S1.

Novel domain identification workflow

Our novel domain identification workflow makes use of several filters that were applied to non–CATH-labeled cluster representatives. First, the representatives of the 41,879,858 domain clusters were assessed using the normalized radius of gyration and packing density described in Bordin *et al.* (6). Globular TED domains were determined with CATH-AlphaFlow as those satisfying both criteria of having a normalized radius of gyration of <0.356 and a packing density of >10.333 (fig. S19). Cutoff values were determined as the 5th percentile of both metrics calculated from the 193,939,492 TED-100 domains with H-level CATH assignments, because these domains were of sufficient quality to be representative of domains in the AFDB.

For secondary structure elements (SSE) filtering, domains were excluded from consideration as novel domains if they contained fewer than six SSEs (helices and strands), as determined by STRIDE (50). STRIDE counts even very short regions (two residues) as separate secondary structures, so a threshold of six elements in total was deemed a sensible threshold to allow us to identify novel domain architectures. Applying both the globularity and SSE filters to the 41,879,858 unlabeled clusters results in 13,820,550 passing both criteria.

Finally, we apply a plDDT filter on the remaining domains, which we call the plDDT_{80} metric. The plDDT_{80} is calculated as the average plDDT of the top 80% of residues ranked by plDDT. We found plDDT_{80} preferable to the simple average because it prevented removing clusters that are mainly well folded but have unfolded tail regions. Applying a $\text{plDDT}_{80} \geq 90$ filter on the remaining 13,820,550 clusters yields a final set of 8,612,318 clusters that pass all three (globularity, SSE, and plDDT_{80}) criteria, encompassing 19,816,697 domains within them.

The filtered sequence cluster representatives are then subjected to layers of further clustering and filtering (Foldseek commands are shown in the supplementary methods). The next step of the workflow involves iterative searches of the remaining cluster representatives against the PDB and CATH (8), ECOD (10), and SCOPe (17) domain libraries using Foldseek (23). These searches use Foldseek's easy search using (exhaustive) TM-align mode, eliminating matches with a TM-align score >0.56 and coverage of 60%. The reason for using a slightly unorthodox TM-align threshold of 0.56 is to compensate for Foldseek approximating the true TM-align score (23), which means that false-positive matches are occasionally found when using the standard threshold of 0.5.

Identifying putative novel folds

Although we consider the remaining 240,674 candidate clusters as our main working set of possible novel domain folds, further processing was performed to try to identify a core subset of domains that were both novel and consistent with the features of well-folded protein domains. First, we separated out the subset of clusters composed of high-symmetry domains. These domains may be composed of known repetitive units, for example, WD40 repeats in beta propellers, but arranged in a manner that confers novelty. Domains with high internal symmetry were identified using the Z score returned by the SymD program (27). We used a threshold of Z score > 9.0 to indicate high-symmetry domains, following the original SymD method paper, which used cutoff values of 8 and 10 (fig. S10).

Detecting properly chopped novel fold domains

Although the use of a consensus domain segmentation method produces generally good-quality domains, inevitably, among the long tail of possible novel folds, we will find candidate domains that represent either over- or underchopped chain segments. By underchopped, we mean structures that still appear to include domain linker regions, and by overchopped, we mean structures that are similar to existing domains but are incomplete, e.g., a segment of a TIM barrel. Thus, we assessed each remaining low-symmetry cluster on chopping quality by retraining a variant of the Foldclass network to identify tenuous domain choppings.

We used a version of the Foldclass network (with three EGNN layers rather than two) to score domains in terms of the quality of the segmentation with a two class (binary) output layer. This new network is trained as a simple binary classifier to distinguish between existing real CATH S30 domains (positives) and random crops of the associated full-length protein chains (negatives). For every true domain, either the true domain is taken with a positive label or a random crop is sampled from the same chain with a negative label. A probability of 0.5 was used for this to ensure a balanced training set. The loss function for training is binary cross entropy, all other training details are the same as for Foldclass. The trained network thus outputs a score between 0 and 1, which indicates the likelihood of a given set of coordinates having been chopped in a manner consistent with the existing CATH domain database. We use a moderate threshold of quality score <0.5 to eliminate any domains that are less likely than not to be correctly chopped, but without rejecting too many clusters that might contain reasonable candidates for globular domains with potentially novel folds. As an additional filter, we also made use of another older domain-parsing algorithm (44) that is generally very good at identifying single-domain proteins based on compactness (domains that were not identified as being single domain with 100% confidence were removed).

The remaining domains were then subjected to increasingly sensitive Foldseek comparisons with the current CATH, ECOD, and SCOPe domain databases (8, 10, 17), with commonly used thresholds of TM-score threshold of 0.5 and query domain coverage of 60%. The TM-score was taken as *max(qtmscore, ttmscore)* so that both the TM scores normalized by the candidate domain length and those normalized by the library domain would be considered. The final stage was to use exhaustive comparisons against the domain libraries using TM-align directly (43). TM-align is generally more sensitive than Foldseek even in its exhaustive mode, and indeed we found further matches (TM-score >0.5 and query coverage >60%) to reduce the final list from 24,653 to 7427 potential novel domain folds. This set of domains represents the set we feel most confident about defining as well-folded domains with no significant structural similarity to current PDB chains (as of December 2023) or current structural domain libraries (as of February 2024).

Using the Foldclass embeddings, we ranked the final list of domains in order of novelty by calculating the mean Euclidean distance between the embedding vector of each candidate domain and the k-nearest neighbors (k = 50) in CATH, ECOD, and SCOPe.

Evaluating domain interactions

To identify which pairs of domain superfamilies interact and the degree of conservation in the interaction geometry for that pair of superfamilies, domains were defined as interacting if there were at least three Cβ atom pairs (Cα in the case of Gly residues) within 8 Å between the two domains. An ISP is the collection of all instances of such contacting pairs of domains belonging to specific superfamilies, and is uniquely identified by the CATH H-level (homologous superfamily) labels comprising it (e.g., 1.10.10.10 to 3.20.20.10). ISPs are identified independently of the order of occurrence of the component domains in the chain sequence and are only defined for domain pairs within the same chain.

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For TED domains, we filter each putative set of interacting domains using the PAE data (2) supplied with each AFDB chain prediction.

Conservation in interaction patterns was assessed using the Conservation of Interaction Orientation (CIO) measure of Littler and Hubbard (45). A CIO score of 0 indicates complete conservation in the interaction geometry (i.e., interaction vectors for a given ISP are absolutely coincident), and a score of 1 indicates a uniformly random angular distribution of vectors. Interaction vectors were only computed for interacting domain pairs as defined above, and all instances of a given ISP are first placed in a common frame of reference by aligning on a common reference structure for one of the domains in the pair using TM-align (43). We also assessed the number of occurrences of ISPs common to both TED and CATH 4.3 expressing relative enrichment as the $\log_2$(fold change) of pairs in TED relative to CATH.

To visualize the sets of ISPs in CATH and TED, we used hierarchical edge bundling plots (46) as implemented in the "ggraph" R package (47), using the CATH hierarchy with a virtual common root for classes 1 to 4 as the guide dendrogram (fig. S13). To identify hub domains, we first take each superfamily and determine the ISPs it is involved in within CATH and TED. A superfamily involved in k ISPs is classified as a "small," "medium," or "large" hub when $k < 4$, $4 \leq k \leq 7$, and $k > 7$, respectively, following the thresholds defined in Ekman *et al.* (48) but applied to domains rather than full-length proteins.

Sequences not modeled in AFDB

The AFDB excludes certain sequences from the UniProt 2021_04 release (https://alphafold.ebi.ac.uk/faq), so these are not covered by TED. To get an idea of the makeup of these sequences, we obtained UniProt release 2021_04 and the "accessions_ids.csv" file for AFDB version 4 from https://ftp.ebi.ac.uk/pub/databases/alphafold/. Using the accessions in the two datasets and limiting ourselves to sequences at least 16 residues long (because these are unlikely to contain nontrivial domains), we find that a total of 10,749,213 sequences do not appear in the AFDB. Using taxonomic data in UniProt, we find that, of these, 5,304,757 are listed as being from viruses. A total of 554,700 and 3,360,117 of these sequences are longer than 2700 and 1280 residues, respectively (up to 2700 residues are allowed for SwissProt or reference proteome sequences, whereas 1280 is the upper limit for other sequences). Although we are unable to provide consensus domain counts for the set of 10.8 million unmodeled sequences because they do not have structures in AFDB, we can estimate domain counts based on the length distribution of these targets. Under the assumption that they exhibit a similar domain count-to-length ratio as the current AFDB sequences, and using TED100 to estimate the number of domains per residue in all modeled UniProt entries, we estimate that there may be ~50 million domains not yet covered by the AFDB, with the caveat that the distribution of non-domain residues could be different for longer sequences.

Validation of homology assignments using hidden Markov models

Consensus domain sequences with an H-level or T-level assignment by Foldseek were scanned against hidden Markov models built from 62,915 CATH representatives clustered at 95% sequence identity. The resulting output from HMMsearch with an E-value cut-off set at 1e-3 were subsequently postprocessed using cath-resolve-hits (49) using a minimum coverage of 80% and minimum bitscore set at 25. Resulting superfamily code assignments for H hits were checked for full matches (identical CATH codes predicted) or fold matches (first three of the CATH digits identical, different fourth-level assignment between the two methods). For T-level assignments by Foldseek, we compared only the first three digits in the CATH code assigned by the HMMs, returning a valid match only if the first three digits are identical, returning a nonmatch otherwise.

Evaluation of domain coverage through sequence and structural searches

Full, nonredundant sequences for AFDB were scanned against hidden Markov models built from 62,915 CATH representatives clustered at 95% sequence identity. The resulting output from HMMsearch with an E-value cut-off set at 1e-3 were subsequently postprocessed using cath-resolve-hits (49) using a minimum coverage of 80% and minimum bitscore set at 25. The tool returns the best nonoverlapping set of domain assignments on each AFDB sequence.

Deriving a consensus domain segmentation

The predicted domains from full-length AFDB targets by Merizo, Chainsaw, and UniDoc were filtered to remove any segments with fewer than five residues and any domains with fewer than 25 residues. To determine a common set of putative domains identified by all three methods, domains are classified into high, medium, or low consensus categories depending on the level of agreement between the predictions of the three methods (fig. S14). Specifically, given a set of N domain predictions for a given target, we construct an N-by-N matrix of pairwise sequence overlap scores between each domain, thresholded at 70% coverage to generate an adjacency matrix. This matrix can be treated as a network graph, with the size of each internal graph component indicative of the confidence level (with 3, 2, and 1 indicating high, medium, and low confidence, respectively).

For TED-100, of which there are 188 million unique sequences, the consensus is derived for each individual protein. High-, medium-, and low-confidence categories reflect the agreement of predictions by either three, two, or single methods, respectively. The final domain ranges reported are taken as the intersection between the regions in agreement with each other. For domain counts and subsequent downstream analysis tasks, only the high and medium categories are considered due to their nonoverlapping nature. Because of the nature of their derivation, high and medium consensus domains cannot overlap.

For TED-redundant, we calculated the consensus domains for the 25,769,418 targets not in TED-100 using the same approach (calculated per chain). Overall, we identified 324,389,697 domains in TED-100 (195,229,271 at high consensus and 129,160,426 at medium consensus) and 40,416,380 domains in TED-redundant (24,482,932 at high consensus and 15,933,448 at medium consensus). The total domain count for TED is taken as the sum of TED-100 and TED-redundant and is 364,806,077 domains, covering all 214,683,829 AFDB targets.

As an alternative view of TED-redundant, we additionally derived a second consensus that takes into account all models for each redundant sequence. This covers all 38,944,835 sequence-redundant targets (including the 13,175,417 representatives part of TED-100). This consensus uses dynamic thresholding for high and medium consensus categories due to the variable number of models being considered for each sequence. Thresholds were derived as $threshold = 3Mk$, where M is the number of targets (two minimum), and k is either 0.66 or 0.33 for high- and medium-confidence levels, respectively. For the TED-redundant multiset only, high-consensus domains are given priority over those of medium consensus, with any in the latter category overlapping with high-consensus domains discarded.

GO term analysis of novel and repeat domains

To try to evaluate the possible functions of our final repeat and novel fold domains, we used a purely sequence-based predictor called GOfocus, which performed well in the 4th CAFA experiment (51) (https://biofunctionprediction.org/cafa/). The model comprises a one-dimensional dilated convolutional network with four blocks of four dilated convolutional layers per block, with dilations cycling through 1, 2, 4, and 8; and a filter size of 3; 512 channels; and residual connections across each layer. After the dilated convolutional blocks, adaptive max-pooling is used to reduce the (L × 512) tensor to fixed dimensions (16 × 512) which are then flattened and used as input features for a two layer fully connected network with ReLU nonlinearity and dropout rate set to 0.1. Sixteen separate networks were trained with different nonoverlapping subsets of a slimmed set of 8360 GO term labels. Subsets were chosen to keep terms with similar

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observed frequencies in the sequence data bank grouped together. Training was performed on the UniRef50 subset of sequences in the December 2019 release of SwissProt and TrEMBL and their respective (non-IEA) GO terms (*52*). Focal loss was used as the loss function and training was performed with the standard Adam optimizer with a learning rate of 1e-4. Testing this model on the CAFA3 benchmark test data produced an $F1_{max}$ score of 0.65 for molecular function terms.

For MMseqs commands used for sequence clustering and Foldseek commands used for novel fold identification workflow, please see the supplementary methods.

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github.com/UCLOrengoGroup/cath-alphaflow). The code for the TED website is available at TED-web (https://github.com/UCLOrengoGroup/ted-web). **License information:** Copyright © 2024 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse. This research was funded in whole or in part by the Wellcome Trust (grant 221327/Z/20/Z).

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SUPPLEMENTARY MATERIALS

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